Point Operations, Histogram Processing, and Pixel Resolution

Digital Image Processing

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Overview

In this project, we explore histogram-based techniques for image contrast enhancement, progressing from basic global transformations to adaptive local methods, and finally extending these concepts to color photography.

According to the assignment requirements, global techniques like Global Histogram Equalization (GHE) are effective for overall contrast adjustment but often fail to preserve small, localized details or can over-amplify noise. To address this, Local Histogram Equalization (LHE) is introduced to adaptively enhance contrast based on local pixel neighborhoods.

Furthermore, applying spatial domain point operations to color images introduces unique challenges. As noted in the assignment, applying Histogram Equalization directly to the R, G, and B channels independently alter the color balance (hue) of the image. To achieve contrast enhancement without color distortion, we implement mathematical color space conversions from scratch ($RGB \leftrightarrow HSV$) and apply equalization strictly to the Value (V) channel, preserving the original chromatic information.

1. Methodology & Approach

1.1 Global Histogram Equalization (GHE)

Global Histogram Equalization applies a single transformation function to the entire image to achieve a uniform distribution of pixel intensities. Let the image have dimensions $M \times N$and discrete intensity levels in the range $[0, L - 1]$(where $L = 256$for 8-bit images).

1. **Probability Distribution:** The probability of occurrence for each intensity level r_k is computed as:

$$P(r_k) = \frac{n_k}{MN}$$

   Where n_k is the total number of pixels with intensity r_k.

2. **Cumulative Distribution Function (CDF):** The discrete transformation function $T(r_k)$ is defined by the CDF:

$$CDF(r_k) = \sum_{j=0}^{k} P\left(r_j\right)$$

3. **Intensity Mapping:** The original pixel values r_k are mapped to the new equalized values s_k using:

$$s_k = \text{round}((L - 1) \times \text{CDF}(r_k))$$

1.2 Local Histogram Equalization (LHE)

To preserve fine, localized details that GHE often washes out, LHE computes the equalization dynamically based on a sliding window.

1. **Neighborhood Definition:** For every pixel at coordinates (x, y), we define a local neighborhood (window) $W_{x,y}$ of size $m \times m$ (e.g. 15×15) centered at (x, y).
2. **Local Transformation:** A local histogram and CDF are computed strictly from the pixels within $W_{x,y}$.
3. **Center Pixel Mapping:** Only the center pixel $(x, y)$is updated based on this local CDF:

$$s_{x,y} = \text{round}((L - 1) \times \text{CDF}_{W_{x,y}}(r_{x,y}))$$

This process is repeated as the window slides across every pixel in the image, allowing the contrast enhancement to adapt to the specific lighting conditions of different regions.

1.3 Color Space Conversion and Equalization

Applying histogram equalization directly to the R, G, and B channels independently alter the relative ratios between the channels, causing hue (color balance) distortion. To prevent this, we isolate the luminance from the chromaticity by converting the image to the HSV color space.

1. **RGB to HSV Conversion (Mathematical Implementation):**
   First, the R, G, and Bvalues are normalized to the range $[0, 1]$. We then find the maximum and minimum color components:
   $$\mathrm{C_{max}} = \max(\mathrm{R, G, B}), \qquad \mathrm{C_{min}} = \min(\mathrm{R, G, B}), \qquad \Delta = \mathrm{C_{max}} - \mathrm{C_{min}}$$
   - **Value(v):** Represents the brightness.
     $$\mathrm{V} = \mathrm{C_{max}}$$
   - **Saturation(S):** Represents color purity.
     $$\mathrm{S} = \begin{cases} 0, & \text{if } \mathrm{C_{max}} = 0 \\ \dfrac{\Delta}{\mathrm{C_{max}}}, & \text{if } \mathrm{C_{max}} \neq 0 \end{cases}$$
   - **Hue (H):** Represents the color type
     $$\begin{cases} 0 & \text{if } \Delta = 0 \\ 60^\circ \times \left(\dfrac{\mathrm{G - B}}{\Delta} \bmod 6\right) & \text{if } \mathrm{C_{max}} = \mathrm{R} \\ 60^\circ \times \left(\dfrac{\mathrm{B - R}}{\Delta} + 2\right) & \text{if } \mathrm{C_{max}} = \mathrm{G} \\ 60^\circ \times \left(\dfrac{\mathrm{R - G}}{\Delta} + 4\right) & \text{if } \mathrm{C_{max}} = \mathrm{B} \end{cases}$$
2. **V-Channel Enhancement:**
   The GHE transformation $T(r)$described in section 2.1 is applied strictly to the Vchannel. The $H$and Schannels remain completely unaltered:
   $$\mathrm{V_{enhanced}} = \mathrm{T(V)}$$

3. **HSV to RGB Reconstruction:**
   The image is converted back to RGB using the modified $V_{enhanced}$ channel alongside the original H and S.

$$\mathrm{C} = \mathrm{V}_{\mathrm{enhanced}} \times \mathrm{S}$$

$$\mathrm{X} = \mathrm{C} \times (1 - \left|(\frac{\mathrm{H}}{60^{\circ}}) \bmod 2 - 1\right|)$$

$$\mathrm{m} = \mathrm{V}_{\mathrm{enhanced}} - \mathrm{C}$$

   Depending on the sector of H (0 to 360 degrees), intermediate values (R', G', B') are assigned combinations of C, X, and 0. Finally, the RGB values have shifted back:

$$(R, G, B) = (R' + m, \qquad G' + m, \qquad B' + m)$$

   By employing this mathematical isolation, we successfully enhance the image's contrast while perfectly preserving its original color balance.

2. Results and Discussion

2.1 Point Operations: Gamma and Piecewise Transformations

Initial experiments applied point-based spatial domain transformations. Gamma correction proved effective at shifting the overall brightness curve—compressing shadows ($\gamma < 1$) or expanding mid-tones ($\gamma > 1$). However, for complex images like the dark moon or hazy trees, a Piecewise-Linear Transformation (Contrast Stretching) provided far superior control. Specifically anchoring control points, we could aggressively stretch the mid-tones while safely compressing the extreme shadows and highlights, resulting in enhanced structural 3D details (like lunar craters) without introducing excessive noise.

2.2 Global vs. Local Histogram Equalization (Grayscale)

Histogram equalization automates contrast enhancement by flattening the intensity distribution.

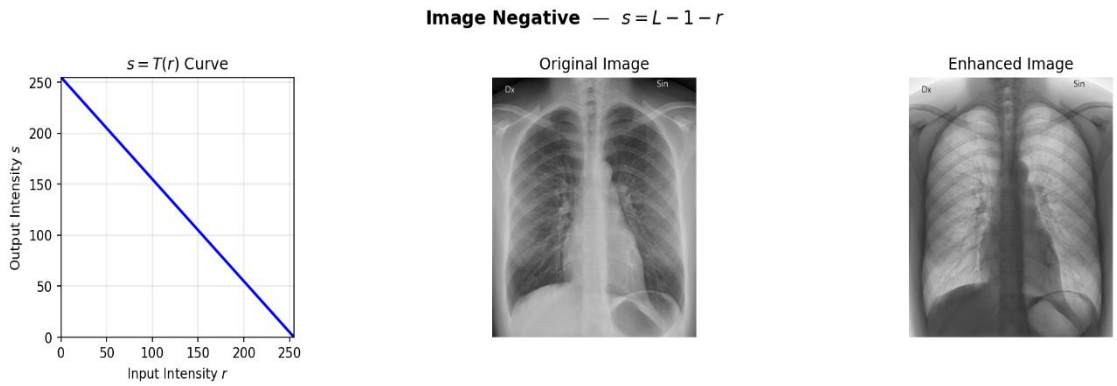
- **Global HE (GHE):** Applied to the entire image, GHE effectively utilized the full $0 - 255$dynamic range. However, because it relies on the macro-distribution of intensities, small, localized details were overshadowed by large background peaks.
- **Local HE (LHE):** By computing localized histograms and CDFs over a sliding 15×15window, LHE successfully revealed hidden micro-structures (such as pulmonary vessels in the chest scan). **Discussion of Trade-offs:** LHE introduced a heavy textured artifact (noise amplification in uniform regions) and suffered from extreme computational cost. Our benchmarking showed LHE took **43.65 seconds** compared to GHE's **0.15 seconds**—an almost 290x increase in execution time due to per-pixel neighborhood recalculations.

2.3 Color Image Enhancements

Applying Histogram Equalization to color images presented unique challenges. When GHE was applied independently to the R, G, and B channels, the relative color ratios were destroyed, resulting in severe color distortion and unnatural hue shifts. By converting the image to the HSV color space (as detailed in Section 1.3) and applying GHE exclusively to the V(Value/Luminance) channel, the image contrast was significantly enhanced while the original H(Hue) and S(Saturation) vectors were mathematically preserved, yielding a natural, vibrant result.

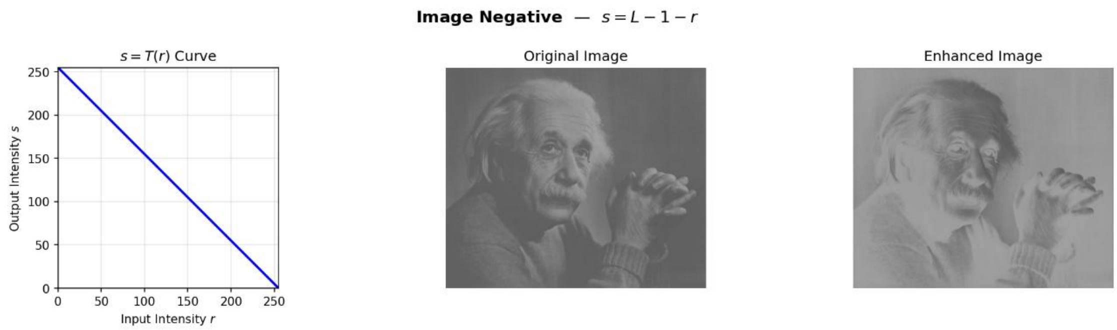
2.1. Image Negative

Medical X-ray (Grayscale)



- **Original:** In standard radiography, denser materials (like bone) absorb more X-rays and appear white, while air and soft tissue appear black or dark gray.
- **Negative Enhanced:** The transformation inverts this relationship. The bones become dark, and the surrounding air/soft tissues become bright white.
- **Purpose/Effect:** This is a very common application of image negatives in medical imaging. Sometimes, the human eye can detect subtle structural details or anomalies more easily when they are presented as dark regions against a light background rather than light regions against a dark background.

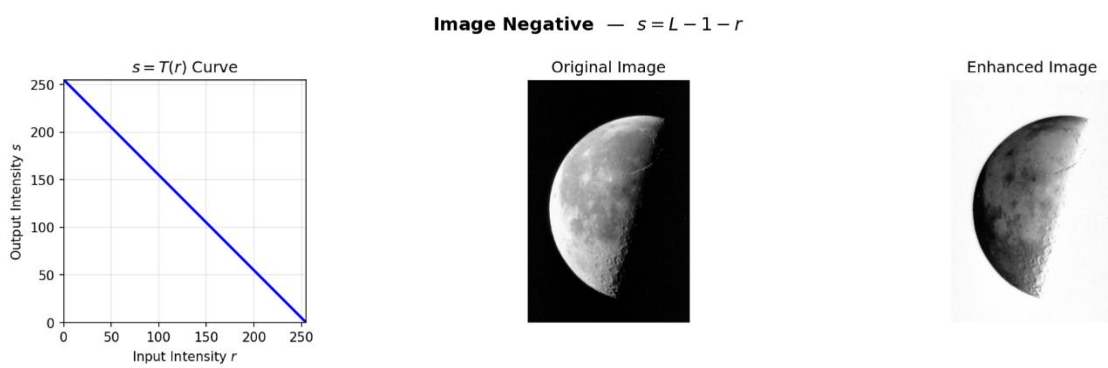
1. **Einstein Portrait (Grayscale)**



- **Original:** A standard photograph with natural lighting, featuring bright highlights on the face and hair, and dark shadows in the background and clothing.

- **Negative Enhanced:** The result perfectly mimics a traditional photographic film negative. The bright white hair becomes pitch black, the shadows on the clothing become glaringly bright, and the mid-tone skin takes on a dark, unnatural appearance. This clearly illustrates how the transformation linearly flips the entire photometric scale of a natural scene.

2. **The Moon (Grayscale)**

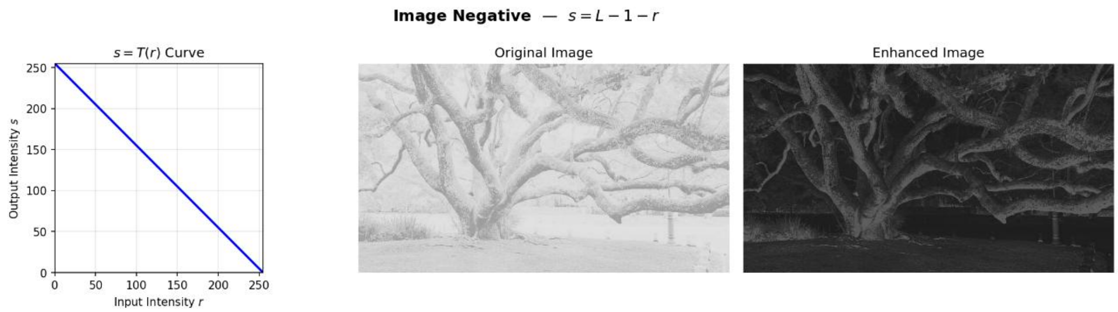


- **Original:** A very high-contrast image consisting of a bright subject (the moon) isolated against an expansive, near-black background (space).

- **Negative Enhanced:** The vast blackness of space is mapped to pure white ($0 \rightarrow 255$). The moon itself becomes a dark sphere. The bright craters and ray systems on the moon's surface become dark etchings.

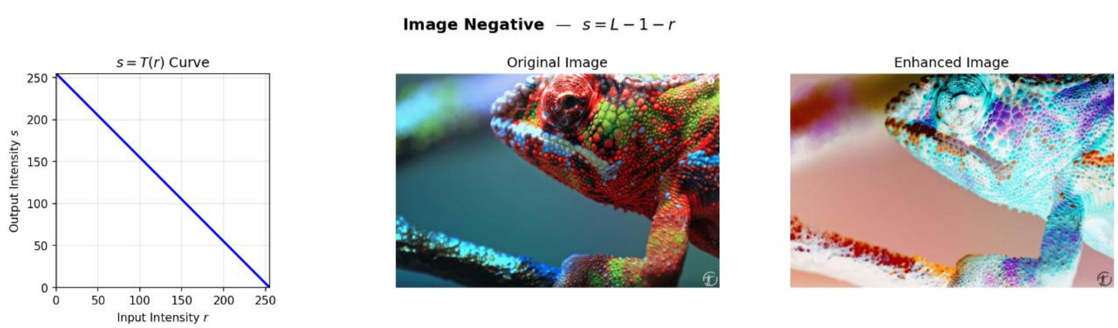
This inversion can sometimes make the faint, dark borders of the moon or subtle surface textures easier to study without the glare of the bright white regions.

Foggy Trees (Grayscale)



- **Original:** A low-contrast, high-key image. The sky and background are very light (foggy), and the tree branches are moderately dark.
- **Negative Enhanced:** The light, foggy background is inverted into a heavy, dark backdrop. The tree branches, which were the darkest elements, are pushed to the highest intensity values, becoming bright white. This completely changes the mood of the image, giving the trees a glowing, skeletal, almost infrared-photography aesthetic.

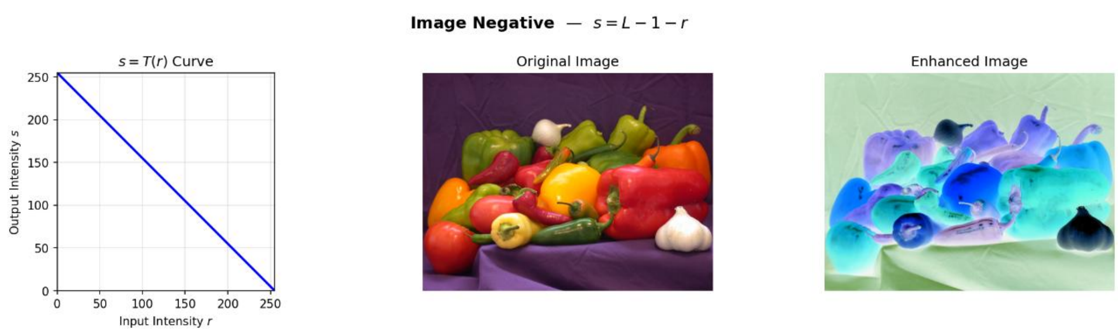
Chameleon (Color)



- **Original:** A nature photograph where the subject and background likely feature natural greens, browns, and yellows.

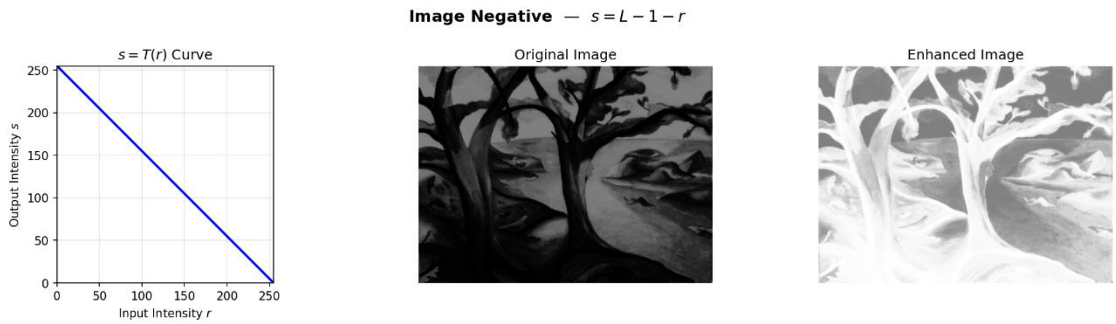
- **Negative Enhanced:** For color images, the $s = 255 - r$operation is applied to the Red, Green, and Blue channels independently. This results in every pixel being replaced by its exact complementary color on the RGB color wheel. Greens become vibrant magentas, yellows become deep blues, and the natural lighting/shadows are completely reversed, giving the chameleon a highly artificial, neon-like appearance.

Peppers (Color)



- **Original:** A brightly colored scene featuring distinct red, green, and yellow peppers against a purple background.
- **Negative Enhanced:** Like the chameleon, the colors are mapped to their complements. The bright red peppers map to a striking cyan. The green peppers turn into purplish magenta. The yellow peppers become blue, and the originally purple background turns into a pale green. This serves as an excellent test case to verify that the RGB channels are being inverted correctly, as the complementary mappings (Red ↔Cyan, Green ↔Magenta, Blue ↔Yellow) are visually obvious.

Dark Landscape Painting (Grayscale)



- **Original:** This image, which appears to be a drawing or painting, is predominantly dark. The tonal range is heavily weighted towards the shadows, with thick, almost black trees in the foreground and dark/mid-gray tones making up the background hills and water.

- **Negative Enhanced:** The tonal hierarchy is completely flipped. The heavy black trees are mapped to pure, glowing white. The darker grays of the background hills are inverted into lighter, hazy gray tones.

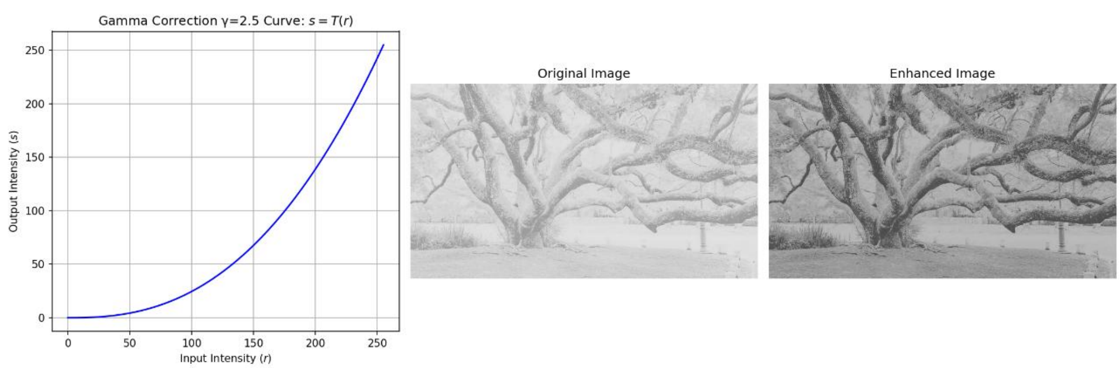
- **Purpose/Effect:** This highlights the artistic and aesthetic impact of the negative transformation. By inverting the image, the previously shadowy silhouettes become the brightest focal points, giving the scene a striking, ethereal, or almost snow-covered winter appearance. It effectively demonstrates how reversing tonal dominance can completely alter the visual mood and emphasize the shapes and outlines of objects that were previously lost in the shadows.

2.2. Gamma Correction (Power-Law Transformation) Analysis

Gamma correction applies to a non-linear power-law transformation defined by the equation $s = c \cdot r^{\gamma}$. By varying the value of γ(gamma), we can selectively compress or expand the dark or bright regions of an image.

Gamma Correction: Low Contrast

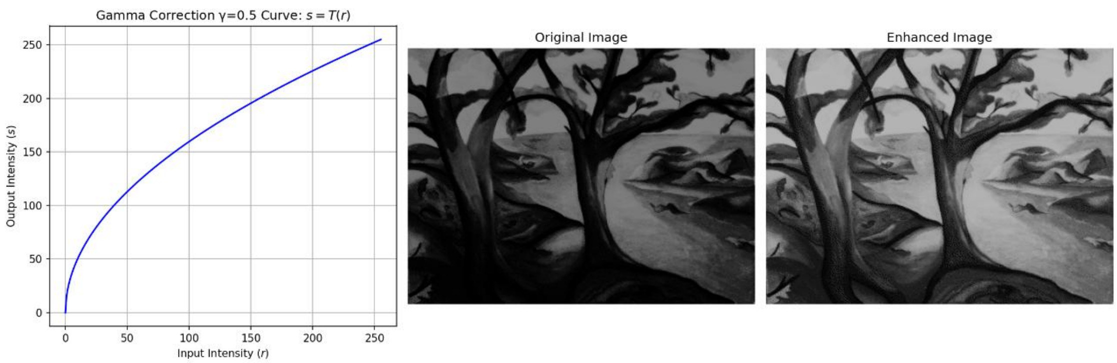
This group demonstrates how gamma correction interacts with an image that lacks dynamic range and has its pixel values predominantly concentrated in the lighter (higher intensity) portion of the histogram, resulting in a faded or foggy appearance.



- **Case C: $\gamma = 2.5$(High Gamma)**
- **Transformation Curve:** The curve bows downward aggressively, mapping a large range of mid-to-light input values to darker output values.
- **Effect on Low-Contrast Image:** For this specific, heavily washed-out image, this higher gamma value acts as a strong corrective measure. It pulls the faded gray branches down into much deeper blacks and dark grays. This creates a strong, almost silhouette-like effect, drastically improving the structural definition of the tree and stripping away the “foggy” look, proving that high gamma values are excellent for recovering detail in overexposed or hazy images.

Gamma Correction Analysis: Dark Landscape Painting

This group demonstrates how gamma correction behaves when applied to a low-key image—an image where the pixel values are predominantly concentrated in the darker (lower intensity) region of the histogram.

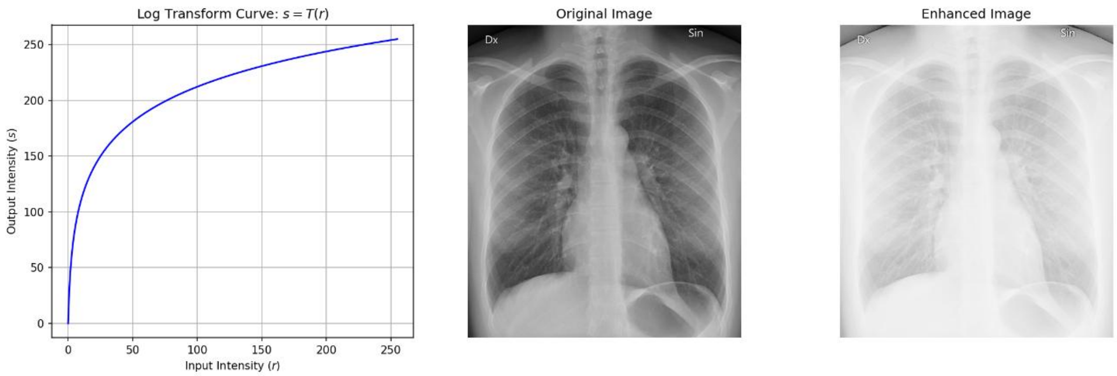


- **Case A:** $\gamma = 0.5$**(Fractional Gamma)**
- **Transformation Curve:** The curve bows upward, aggressively mapping lower input intensities to significantly higher (brighter) output values.
- **Effect on Dark Image:** For an image that is originally very dark and filled with heavy shadows, a fractional gamma is highly beneficial. The transformation lifts the dark grays and blacks, revealing details in the trees and the terrain that were previously hidden in the shadows. While it slightly flattens the overall contrast, it successfully makes the scene much more visible and comprehensible.

2.3. Logarithmic Transformation

Logarithmic Transformation Analysis: Medical X-ray

This example explores the application of the Log Transformation, governed by the general formula $s = c\log{(1 + r)}$, to a chest radiograph.

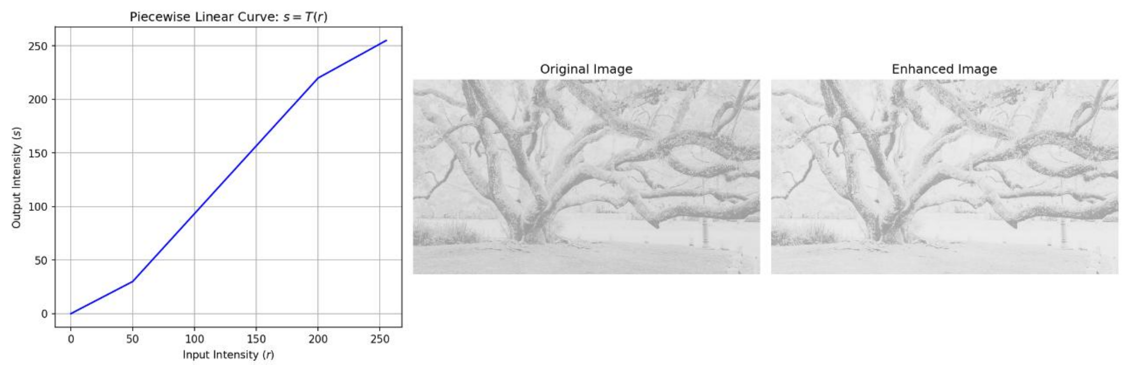


- **Transformation Curve:** The plot illustrates the characteristic shape of the log function. It rises very steeply at the lower input intensities (0 to ≈ 50) and then severely flattens out for higher input intensities. This means that a narrow range of dark input values is mapped to a very wide range of brighter output values, while the bright input values are compressed into a very narrow range at the top end of the scale.
- **Effect on Medical Image:** The original X-ray relies heavily on the stark contrast between radiolucent areas (like air in the lungs, appearing dark) and radiopaque structures (like bones, appearing bright).
- Because the log transform aggressively expands dark values, the dark regions of the lungs and the black background are drastically lightened, turning into pale grays.
- Conversely, the brighter pixels of the ribcage and spine are compressed, losing their subtle textural differences.
- **Result:** The resulting enhanced image is severely washed out and overexposed. The critical diagnostic contrast between the soft tissues/air and the skeletal structure is virtually destroyed.

2.4. Piecewise-Linear Contrast Stretching Analysis

This section explores the effect of a piecewise-linear contrast stretching transformation. For these examples, the transformation function $s = T(r)$is defined by two control points: $(r_1, s_1) = (50{,}30)$and $(r_2, s_2) = (200{,}220)$. Analyzing the slopes of this specific configuration reveals its purpose:

Foggy Trees (Mild Stretch)



- Transformation Parameters: This transformation uses the control points $(r_1, s_1) = (50{,}30)$and $(r_2, s_2) = (200{,}220)$. This yields a slope of approximately 1.26in the mid-tones, while the shadows and highlights are compressed with slopes less than 1(≈ 0.6 and ≈ 0.63, respectively).
- Original State: The image is very low contrast, hazy, and pale. It is a "high-key" image, meaning most of its pixel intensities are concentrated in the lighter gray (higher intensity) ranges due to the thick fog. The darkest parts of the tree branches are only medium gray.
- Enhancement Effect:
- Mid-tones: Because most of the structural detail (the branches against the fog) falls within the $50$to 200input range, the mild stretch (slope $\approx$ 1.26) gently pulls these values apart. This slightly darkens the branches while brightening the brighter patches of fog.
- Highlights: The upper segment compresses the brightest values, which ensures the fog doesn't completely "blow out" to a featureless pure white.
- Result: The resulting enhancement is very subtle. There is a slight improvement in contrast, bringing out a bit more definition in the darker tree bark and branches. Crucially, because the stretch is mild, the

transformation successfully retains the original hazy, atmospheric, high-key aesthetic of the scene without introducing harsh, unnatural transitions.

2.5.Histogram Calculation & Plotting

To establish the foundation for histogram-based contrast enhancement, a custom function was developed to compute the histogram $h(r_k)$and the normalized Cumulative Distribution Function (CDF), $p(r_k)$, directly from the image array without relying on built-in functions like cv2.calcHist.

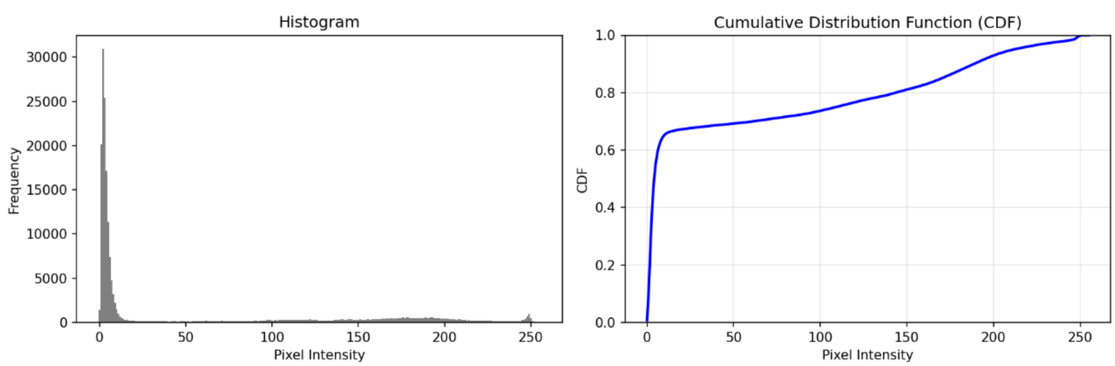
The algorithm iterates through the image pixels, counting the frequency of each intensity level $r_k \in [0,255]$to build the histogram. The normalized CDF is then computed using the cumulative sum of these probabilities, defined mathematically as:

$$s_k = T(r_k) = \sum_{j=0}^{k} p(r_j) = \sum_{j=0}^{k} \frac{n_j}{N}$$

where $n_j$is the number of pixels with intensity r_j, and $N$is the total number of pixels in the image.

Analysis of the moon.tif Test:

Applying this custom function to the moon.tif image yields highly characteristic plots:



- **Histogram Analysis:** The histogram features a massive, narrow peak at the very low-intensity range (near 0). This perfectly corresponds to the vast, empty black space surrounding the moon, which dominates the total pixel count. The actual surface details of the moon are represented by the very short, spread-out frequencies spanning the mid-to-high intensity ranges.

- **CDF Analysis:** Consequently, the CDF curve exhibits an extremely sharp, nearly vertical initial slope, jumping from $0$to roughly 0.65almost immediately. This indicates that roughly 65% of the image consists of the dark background. After this sharp rise, the CDF transitions into a gentle, gradual slope as it accumulates the scattered mid-to-high intensity pixels of the moon itself, eventually reaching 1.0.

This highly skewed distribution anticipates the challenges of applying Global Histogram Equalization (GHE) to such an image, as forcing a uniform distribution on this data will likely stretch the massive dark background into lighter gray tones.

2.6. Global Histogram Equalization (GHE)

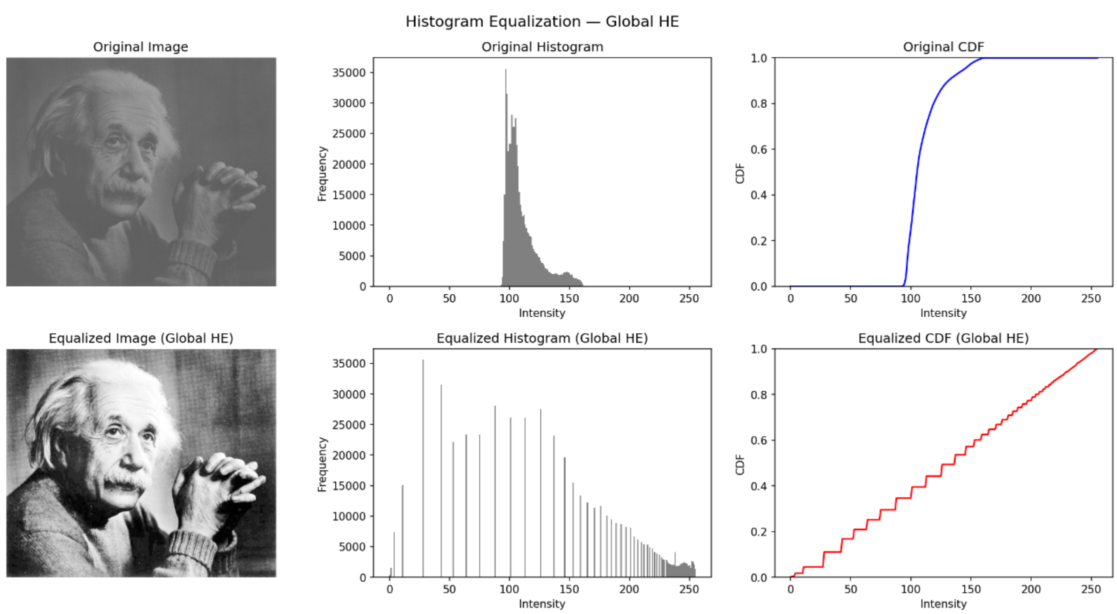
Global Histogram Equalization utilizes the normalized Cumulative Distribution Function (CDF), calculated in the previous step, as a transformation function to remap the intensity values of the entire image. The goal is to produce an output image with a more uniform (flat) histogram, effectively stretching the contrast over the entire available dynamic range. For an L-level image (e.g., $L = 256$for 8-bit), the transformation is applied as:

$$s_k = \mathrm{round}((L-1) \times p(r_k))$$

where $r_k$is the input intensity and $s_k$is the output equalized intensity.

Analysis of the Global HE Results:

To demonstrate this, GHE was applied to a low-contrast image of Albert Einstein.



- **Original Image & Distribution:** The original image appears heavily "washed out" with very poor contrast. Examining its histogram confirms this visual assessment: the pixel intensities are tightly clustered in a narrow band of mid-gray values (roughly between $100$and 150).

Consequently, the original CDF shows a steep, sudden rise strictly within this narrow interval and is completely flat everywhere else.

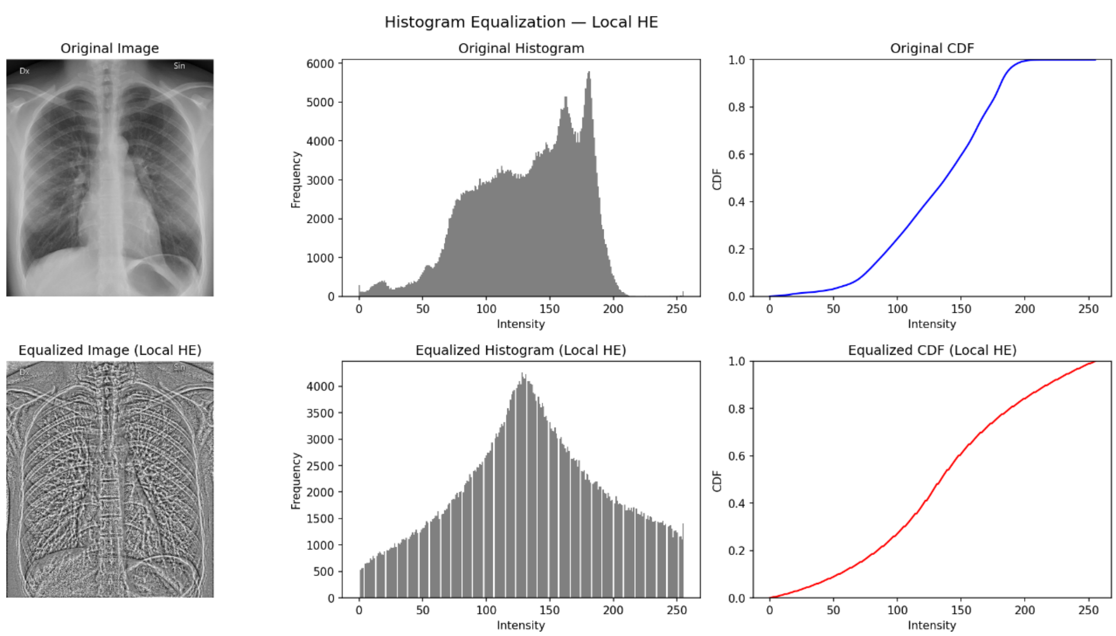
- **Equalized Image & Distribution:** After applying the GHE mapping, the global contrast of the image is drastically improved, revealing facial features and background textures much more clearly.
- **Histogram Flattening:** The equalized histogram demonstrates that the tight cluster of intensities has been successfully stretched across the full $0$to 255dynamic range. Note the presence of “gaps” or zero-frequency bins in the equalized histogram; this is a standard characteristic of discrete histogram equalization, as pixels sharing the exact same input intensity cannot be split into different output bins.
    - **Linearized CDF:** The equalized CDF now closely approximates a straight diagonal line. While it exhibits a “stair-step” pattern due to the discrete nature of digital images, this linear trend indicates that the intensity distribution has been successfully forced toward a much flatter, more uniform state.

2.7. Local Histogram Equalization (LHE)

While Global Histogram Equalization (GHE) is highly effective at improving overall image contrast, it bases its transformation on the intensity distribution of the *entire* image. Consequently, it often fails to enhance—and can even wash out—small, localized details if their local intensity distributions are overshadowed by massive global peaks (e.g., large background areas).

To address this, Local Histogram Equalization (LHE) applies the equalization process on a neighborhood basis. A sliding window (e.g., 15×15pixels) is moved across the image. For every single pixel, a local histogram and local CDF are computed strictly from the pixels within that window. The center pixel is then mapped to its new value using this local CDF, and the window moves to the next pixel.

Analysis of LHE Results (Chest X-Ray):

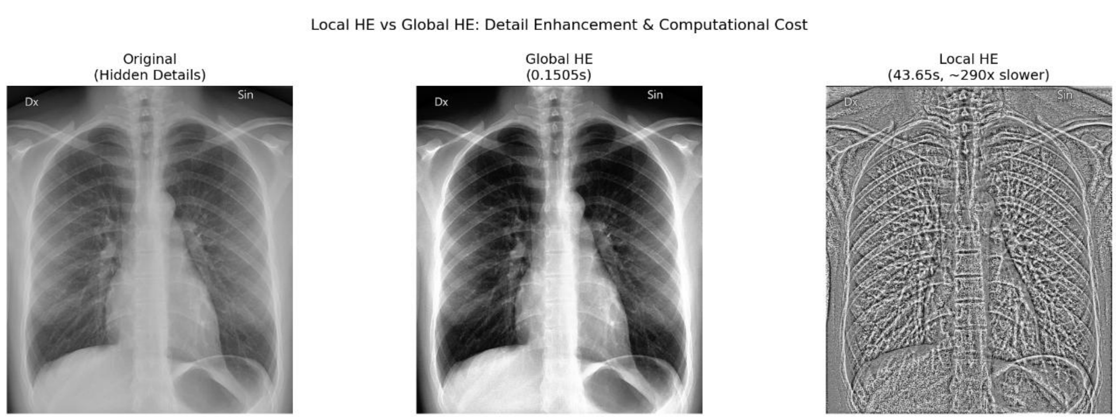


The application of LHE to the medical X-ray image provides a stark contrast to the GHE approach, highlighting three major discussion points:

1. **Revelation of Hidden Details:** The side-by-side comparison clearly shows LHE's superiority in detail extraction. While GHE simply darkened the lungs and brightened the ribs, LHE adapted to the local neighborhoods. This localized stretching forced the hidden micro-structures—such as the fine pulmonary vessels, tissue textures, and overlapping bone structures—to become hyper-visible.

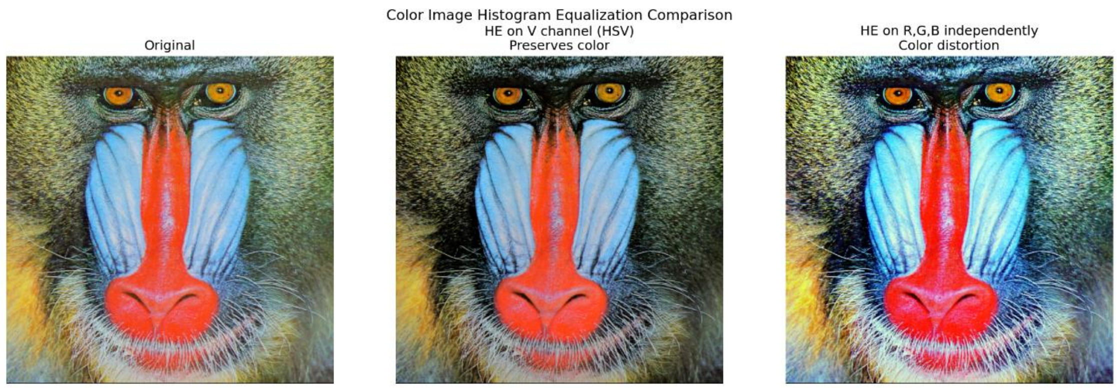
2. **Noise Enhancement and Artifacts:** The heavy extraction of detail comes at a cost. Because LHE forces a full-range contrast stretch in *every* small 15×15window, it heavily amplifies background noise and small variations in otherwise uniform regions. The resulting LHE image looks highly textured, almost embossed, losing the natural global shading and lighting of the original radiograph.

3. **Computational Cost:** The most significant drawback of LHE is its extreme computational expense. As shown in the side-by-side figure, GHE processed the image in just **0.15 seconds**, requiring only one histogram computation. LHE, however, took **43.65 seconds** (roughly 290 times slower). This massive delay occurs because the algorithm must compute a brand-new histogram, calculate a new CDF, and perform a transformation mapping for *every single pixel* in the image based on its surrounding neighborhood.



2.8. Extension: Histogram Equalization on Color Images

While Histogram Equalization (HE) is straightforward for grayscale images, applying it to color images requires careful handling of color spaces to avoid unwanted artifacts. To demonstrate this, global HE was applied to a standard color test image using two different methodologies.



1. Independent RGB Equalization (Color Distortion):

The naive approach is to separate the image into its Red, Green, and Blue (RGB) channels, apply histogram equalization to each channel independently, and then merge them back together. As shown in the third panel of the results, this causes severe color shifting and distortion. Because each color channel has a different initial intensity distribution (a different histogram), equalizing them independently changes the relative ratios of R, G, and B for any given pixel. Since the ratio of these values defines the specific color (hue), the original colors are destroyed, resulting in an unnatural, over-saturated appearance.

2. HSV V-Channel Equalization (Color Preservation):

The correct approach to enhancing the contrast of a color image without altering its original colors is to decouple the intensity (luminance) from the color information (chrominance).

To achieve this, the image was first converted from the RGB color space to the HSV (Hue, Saturation, Value) color space. The Histogram Equalization algorithm was then applied **only to the V (Value/Intensity) channel**, leaving the Hue and Saturation channels completely untouched. Finally, the image was converted back to RGB.

As seen in the middle panel of the results, this method successfully enhances the overall contrast and brightness of the image while perfectly preserving the natural colors of the original image.

Conclusion

This report explored various spatial domain image enhancement techniques, ranging from fundamental point operations to advanced statistical histogram manipulations. Point operations like Piecewise-Linear transformations offer the highest degree of manual control, making them ideal for targeted enhancement.

Histogram Equalization provides a powerful, automated approach to contrast enhancement. While Global HE is computationally efficient and excellent for overall brightness adjustment, it fails to enhance localized, low-contrast details. Local HE solves this limitation by adapting to local statistical distributions, but it does so at the cost of intense computational overhead and potential noise amplification. Finally, we demonstrated that spatial transformations on color images must be decoupled from chrominance; isolating luminance via the HSV color space is critical to preserving the true color integrity of an image during contrast stretching.